

BioTwin: AI-Powered Digital Health Twin for Predictive Healthcare

BYOP 2026 Submission | IIT Roorkee

Vision

BioTwin aims to develop an AI-driven Digital Health Twin that creates a dynamic virtual representation of an individual's health. It integrates lifestyle data, clinical reports, and biomarker information to enable early prediction of diseases and personalized preventive healthcare.

The objective is to shift healthcare from a reactive model to a predictive and preventive intelligence system.

PROBLEM STATEMENT

Current Problem

Modern healthcare systems are largely reactive:

- Diseases are detected only after symptoms appear
- No continuous personalized monitoring exists
- Health data is fragmented across reports, devices, and habits

Key Gaps

- Lack of unified health data integration
- No predictive modeling for long-term health risks
- Limited personalization in preventive healthcare systems

Why This Matters

- Rising chronic diseases (diabetes, cardiovascular disorders)
- Lack of early warning systems for individuals
- Increasing need for personalized healthcare solutions

PROPOSED SOLUTION

BioTwin System Overview

BioTwin creates a digital health twin by integrating multiple data sources and applying AI models for prediction and recommendation.

System Modules

1. Data Collection Layer

- Clinical reports
- Lifestyle data (diet, sleep, activity)
- Biomarker inputs

2. Data Processing Layer

- Cleaning and normalization
- Feature extraction

3. Digital Twin Engine

- Constructs personalized health profile

4. AI Prediction Model

- Risk prediction for diseases
- Trend analysis of biomarkers

5. Recommendation System

- Personalized health suggestions
- Preventive care guidance

6. Dashboard Interface

- Visual health score
- Risk visualization
- Progress tracking

Core Idea

A continuously evolving AI system that models an individual's health trajectory and predicts future risks before clinical

ARCHITECTURE & TECHNOLOGY

System Architecture

User Data → Processing Layer → Digital Twin
Engine → AI Prediction Model →
Recommendation System → Dashboard

Technology Stack

- Python
- Machine Learning Algorithms
- Streamlit / Web Framework
- Large Language Models (LLMs)
- Data Visualization Libraries
- SQLite / Database Systems

Key Features

- Real-time health profiling
- Risk prediction system
- Biomarker trend analysis
- Personalized recommendations
- Interactive dashboard

IMPACT & FUTURE SCOPE

Expected Impact

- Early detection of chronic diseases
- Personalized preventive healthcare
- Improved health awareness and decision-making
- Reduction in long-term healthcare risks

Future Scope

- Integration with wearable devices (smartwatches, fitness trackers)
- Genomic and multi-omics data integration
- Real-time health monitoring system
- Clinical deployment for hospitals
- AI-driven precision medicine platform

Novelty Statement

Unlike traditional health applications, BioTwin creates a continuously evolving digital health twin that models long-term health trajectories using AI-based predictive analytics.

Conclusion

BioTwin represents a step toward intelligent, preventive healthcare systems that combine artificial intelligence with biomedical science to transform how health is monitored and managed.